



Savitribai Phule Pune University
Fourth Year of Computer Engineering (2019 Course)
410253A: Elective-VI Business Intelligence

Teaching Scheme: TH: 03 Hours/Week	Credit: 03	Examination Scheme: Mid-Semester (TH) : 30 Marks End-Sem (TH): 70 Marks
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Prerequisites Courses: 310241: Database Management System, 310251: Data Science & Big data Analytics, 410242: Machine Learning

Companion Course: 410256: Laboratory Practice VI

● **Course Objectives:**

1. To introduce the concepts and components of Business Intelligence (BI)
2. To evaluate the technologies that make up BI (data warehousing, OLAP)
3. To identify the technological architecture of BI systems
4. To explain different data preprocessing techniques
5. To identify machine learning model as per business need
6. To understand the BI applications in marketing, logistics, finance and telecommunication sector

● **Course Outcomes:** On completion of this course, the students will be able to

CO1: Differentiate the concepts of Decision Support System & Business Intelligence

CO2: Use Data Warehouse & Business Architecture to design a BI system.

CO3: Build graphical reports

CO4: Apply different data preprocessing techniques on dataset

CO5: Implement machine learning algorithms as per business needs

CO6: Identify role of BI in marketing, logistics, and finance and telecommunication sector

Course Contents

Unit I	Introduction to Decision support systems and Business intelligence	07 Hours
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Decision support systems: Definition of system, representation of the decision-making process, evolution of information systems, Decision Support System, Development of a decision support system, the four stages of Simon's decision-making process, and common strategies and approaches of decision makers

Business Intelligence: BI, its components & architecture, previewing the future of BI, crafting a better experience for all business users, End user assumptions, setting up data for BI, data, information and knowledge, The role of mathematical models, Business intelligence architectures, Ethics and business intelligence

#Exemplar/Case Studies	Decision support system in business intelligence: https://www.riverlogic.com/blog/five-decision-support-system-examples
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*Mapping of Course Outcomes for Unit I	CO1
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Unit II	The Architecture of DW and BI	07 Hours
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BI and DW architectures and its types - Relation between BI and DW - OLAP (Online analytical processing) definitions - Different OLAP Architectures-Data Models-Tools in Business Intelligence-Role of DSS, EIS, MIS and digital Dash boards – Need for Business Intelligence
 Difference between OLAP and OLTP - Dimensional analysis - What are cubes? Drill-down and roll-up - slice and dice or rotation - OLAP models - ROLAP versus MOLAP - defining schemas: Stars, snowflakes and fact constellations.

#Exemplar/Case Studies	A case study on Retail Industry : https://www.diva-portal.org/smash/get/diva2:831050/FULLTEXT01.pdf	
*Mapping of Course Outcomes for Unit II	CO2	
Unit III	Reporting Authoring	07 Hours
Building reports with relational vs Multidimensional data models; Types of Reports – List, crosstabs, Statistics, Chart, map, financial etc; Data Grouping & Sorting, Filtering Reports, Adding Calculations to Reports, Conditional formatting, Adding Summary Lines to Reports. Drill up, drill- down, drill-through capabilities. Run or schedule report, different output forms – PDF, excel, csv, xml etc.		
#Exemplar/Case Studies	Power BI Case Study – How the tool reduced hassles of Heathrow & Edsby: https://data-flair.training/blogs/power-bi-case-study/	
*Mapping of Course Outcomes for Unit III	CO3	
Unit IV	Data preparation	07 Hours
Data validation: Incomplete data , Data affected by noise . Data transformation: Standardization , Feature extraction. Data reduction : Sampling, Feature selection, Principal component analysis, Data discretization . Data exploration : 1.Univariate analysis :Graphical analysis of categorical attributes ,Graphical analysis of numerical attributes , Measures of central tendency for numerical attributes , Measures of dispersion for numerical attributes, Identification of outliers for numerical attributes 2.Bivariate analysis: Graphical analysis , Measures of correlation for numerical attributes , Contingency tables for categorical attributes, 3.Multivariate analysis: Graphical analysis , Measures of correlation for numerical attributes		
#Exemplar/Case Studies	Case study on Data preparation phase of BI system https://blog.panoply.io/load-and-transform-how-to-prepare-your-data-for-business-intelligence	
*Mapping of Course Outcomes for Unit IV	CO4	
Unit V	Impact of Machine learning in Business Intelligence Process	07 Hours
Classification: Classification problems, Evaluation of classification models, Bayesian methods, Logistic regression. Clustering: Clustering methods, Partition methods, Hierarchical methods, Evaluation of clustering models. Association Rule: Structure of Association Rule, Apriori Algorithm		
#Exemplar/Case Studies	Business applications for comparing the performance of a stock over a period of time https://cleartax.in/s/stock-market-analysis	
*Mapping of Course Outcomes for Unit V	CO5	
Unit VI	BI Applications	07 Hours

Tools for Business Intelligence, Role of analytical tools in BI, Case study of Analytical Tools: WEKA, KNIME, Rapid Miner, R; Data analytics, Business analytics, ERP and Business Intelligence, BI and operation management, BI in inventory management system, BI and human resource management, BI Applications in CRM, BI Applications in Marketing, BI Applications in Logistics and Production, Role of BI in Finance, BI Applications in Banking, BI Applications in Telecommunications, BI in salesforce management	
#Exemplar/Case Studies	Logistics planning in the food industry https://www.foodlogistics.com/case-studies https://www.barrettdistribution.com/food-distribution-case-study
*Mapping of Course Outcomes for Unit VI	CO6

Learning Resources

Text Books:

1. Fundamental of Business Intelligence, Grossmann W, Rinderle-Ma, Springer, 2015
2. R. Sharda, D. Delen, & E. Turban, Business Intelligence and Analytics. Systems for Decision Support, 10th Edition. Pearson/Prentice Hall, 2015

Reference Books :

1. Paulraj Ponnian, "Data Warehousing Fundamentals", John Willey.
2. Introduction to business Intelligence and data warehousing, IBM, PHI
3. Business Intelligence: Data Mining and Optimization for Decision Making, Carlo Vercellis, Wiley, 2019
4. Data Mining for Business Intelligence, WILEY
5. EMC Educational Services, Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data, Wiley ISBN-13 978 1118876138
6. Ken W. Collier, Agile Analytics: A value driven Approach to Business Intelligence and Data
7. Warehousing, Pearson Education, 2012, ISBN-13 978 8131786826

e-Books :

1. https://www.knime.com/sites/default/files/inline-images/KNIME_quickstart.pdf
2. www.cs.ccsu.edu/~markov/weka-tutorial.pdf
3. http://www.biomedica-help.altervista.org/Magistrato/Clinics/BIC_PrimoAnno/IdentificazioneModelliDataMining/Business%20Intelligence%20-%20Carlo%20Vercellis.pdf
4. <https://download.e-bookshelf.de/download/0000/5791/06/L-G-0000579106-0002359656.pdf>

NPTEL/YouTube video lecture links:

- Business Analytics for management decision : <https://nptel.ac.in/courses/110105089>
- Business analytics and data mining modeling using R : <https://nptel.ac.in/courses/110107092>
- Business Analysis for Engineers : <https://nptel.ac.in/courses/110106050>

@The CO-PO Mapping Matrix

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	2	1	1	2	-	-	-	-	-	-	-
CO2	1	1	1	1	1	-	-	-	-	-	-	-
CO3	1	2	1	1	1	-	-	-	-	-	-	-
CO4	2	2	2	1	1	-	-	-	-	-	-	-
CO5	2	2	2	2	1	-	-	-	-	-	-	-
CO6	-	1	-	1	1	-	-	-	-	-	-	-